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M.S. Student

Ulsan National Institute of Science and Technology (UNIST)

Research Interests

- Brain-Computer Interfaces / Brain-Machine Interfaces
- Neural Encoding & Decoding of Vision / Motor Control
- Cognitive Neuroscience of Sensory Perception
- Brain Imaging

Skills

Tools: MATLAB, Python, EEGLAB, SPM, FSL, MySQL

Experiments: EEG, fMRI, Eye-tracking

Education

2025 – Current	M.S. Biomedical Engineering – BCI LAB (Prof. Sung-Phil Kim) Ulsan National Institute of Science and Technology (UNIST), Ulsan, South Korea
2019 – 2025	B.S. Biomedical Engineering / Electrical Engineering Ulsan National Institute of Science and Technology (UNIST), Ulsan, South Korea
2023 – 2024	Republic of Korea Air Force / Tactical Air Control Party – Air Support Operations Center (Mandatory Military Service)
2016 – 2018	Hansung Science High School, Seoul, South Korea

Research Experience

2020 – Current	<u>Brain-Computer Interface Lab</u> , UNIST (Advisor: Prof. Sung-Phil Kim) ▪ Led human neurophysiology studies using EEG (and EEG + eye-tracking) with end-to-end responsibility from experimental design to analysis (MATLAB/EEGLAB; Python as needed). ▪ Key contributions across projects: paradigm development and implementation, participant recruitment/running sessions, preprocessing/feature extraction, and interpretation/reporting of results.
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[Selected Projects]

- **EEG responses of emotion for various subscription fees on OTT services**
(Joint project with Korea University)

Designed and implemented the experimental paradigm; conducted EEG experiments with 10+ participants; analyzed EEG data to characterize emotional differences across cost conditions.

- **EEG responses of choice overload effect**

Developed and executed a complex EEG + eye-tracking paradigm; conducted experiments with 50+ participants; processed/analyzed large-scale datasets to probe neural processes under different preference conditions.

▪ **fMRI study on tactile illusion**

Conducted 3T fMRI experiments with 30+ participants; analyzed BOLD signals using MATLAB/SPM/FSL; compared EEG vs fMRI methodologies for sensory mechanism studies. ▪ EEG signal analysis using EEGLAB and MATLAB.

Spring 2022

Decision Neuroscience and Cognitive Engineering Lab, UNIST (Advisor: Prof. Dongil Chung)

▪ **Resting-State EEG Analysis of Drug-Induced Parkinsonism and Parkinson's Disease**

(Joint project with Dongtan Sacred Heart Hospital)

Analyzed resting-state EEG from PD and DIP patients; performed spectral analyses and EEG-clinical correlation analyses to identify differentiating neural signatures and support diagnostic insight.

Summer 2020

Bio and Neural Interfaces Lab / KAIST / Advisor: Prof. Seongjun Park

▪ **Mechanical Interaction of Thermally Drawn Fiber Implants with Brain Tissue**

Investigated tissue-device interactions for fiber-based neural implants using in vivo surgery, histology, and computational modeling.

Performed mouse brain implant surgeries with fiber-based neural probes; verified immune responses via immunohistochemistry; conducted finite element analysis to model mechanical interactions between probes and brain tissue.

Publications

- [1] **S. Seo***, S. Lee*, N. Lee & S.-P. Kim. (2026). Effects of Cognitive Demand Reduction on Choice Overload. *bioRxiv*, 2026-02.
- [2] **S. Seo***, S. Kim*, S.-P. Kim, J.-H. Kim, S. Y. Kang & D. Chung. (2024). Low-frequency EEG power and coherence differ between Drug-induced Parkinsonism and Parkinson's Disease. *Clinical Neurophysiology*, 168, 131-138.
- [3] J. Park, S. Wang, S. Lee, **S. Seo**, N. Lee & S. Kim. (2023). Viewer Emotional Response to Webtoon-Based Drama: An EEG Analysis. *International Journal of Human-Computer Interaction*, 1-15.
- [4] K. Kim, C. Sung, J. Lee, J. Won, W. Jeon, **S. Seo**, K. Yoon & S. Park. (2021). Computational and Histological Analyses for Investigating Mechanical Interaction of Thermally Drawn Fiber Implants with Brain Tissue. *Micromachines*, 12(4), 394.
- [5] S. Lee, **S. Seo**, N. Lee & S.-P. Kim. (2021). The Effect of OTT Subscription Fee on Content Engagement. *The Journal of Korean Institute of Communications and Information Sciences*, 46(3), 572-582.

Conference Presentations

- [1] **S. Seo**, O.-S. Kwon & S.-P. Kim. (May 2026). EEG Neurometric Function Mirrors Psychometric Performance in Visual Motion Perception. *Vision Sciences Society 2026*. St. Pete Beach, FL.
- [2] J. Kim, C.-E. Yoon, **S. Seo** & S.-P. Kim. (Feb 2026). Real-Time Single-Trial Decoding in a Calibration-Free P300-Based BCI Using a Compact Eight-Channel EEG Configuration. *2026 14th International Conference on Brain-Computer Interface (BCI)*. Jeongseon, South Korea.
- [3] **S. Seo***, S. Kim, SP. Kim, J. Kim, D. Chung & SY. Kang. (Sep 2024). Quantitative EEG Analysis for Comparing

Drug-Induced Parkinsonism and Parkinson's Disease. *International Congress of Parkinson's Disease and Movement Disorders*. Philadelphia, PA.

- [4] J.-H. Kim, J. Kim, **S. Seo** & S.-P. Kim. (Nov 2022). Somatosensory cortical representations of assimilation effects by vibrotactile stimulation: an fMRI study. *Society for Neuroscience 2022*. San Diego, CA.
- [5] **S. Seo**, S. Lee, N. Lee & S.-P. Kim. (Sep 2022). Preference modulates the effect of choice overload with less involvement of working memory. *International Congress of Physiological Anthropology 2022*. Eugene, OR.
- [6] S. Lee*, **S. Seo***, N. Lee & S.-P. Kim. (May 2022). Attenuated working memory accounts for choice overload in highly preferred alternatives. *The 25th Annual Meeting of the Korean Society for Brain and Neural Sciences*. Incheon, South Korea.

Research Grants

- [1] Augmentation Technology for Overcoming the Limits and Disorders of Human Cognition and Intelligence, **Graduate researcher**, 2026.03 – Current
Information Technology and Telecommunications Planning and Evaluation Institute, 2024.07 – 2031.12
- [2] Music BrainCoder: Development of a Non-Invasive Brain-Computer Interface for Reproducing Imagined Music and Enhancing Musical Performance, **Graduate researcher**, 2025.09 – Current
National Research Foundation of Korea, 2025.03 – 2030.02
- [3] Future Brain Convergence Technology Development Project, **Undergraduate researcher**, 2022.08 – 2023.02
“Development of brain-inspired computational models of active and adaptive intelligence for AI applications”
Ministry of Science and ICT, 2019.06 – 2023.12
- [4] Brain Science Proprietary Technology Development Project, **Undergraduate researcher**, 2022.04 – 2023.02
“Development of the analysis methods of ultra-high-resolution neuroimaging signals and the information processing models of neural networks”
Ministry of Science and ICT, 2019.06 – 2023.12
- [5] Smart Healthcare Research Center, **Undergraduate researcher**, 2022.03 – 2022.07
- [6] Smart Media Services Innovation Convergence Project, **Undergraduate researcher**, 2020.07 – 2022.02
“Development of Neuro-Artificial Intelligence Convergence Technology for Next Generation OTT Service”
Ministry of Science and ICT, 2020.07 – 2027.12

Honors & Awards

2022	Excellent Research Award, 2nd place The Faculty of the Department of Biomedical Engineering, UNIST, South Korea
2021	Student Creative Research Project, 1st place Smart media Service Research Center, Korea University, South Korea
2020	Outstanding Grades for the Semester (Article 62 of the School Regulations) UNIST, South Korea
2020	Student Creative Research Project, 2nd place Smart media Service Research Center, Korea University, South Korea
2019	Outstanding Grades for the Semester (Article 62 of the School Regulations) UNIST, South Korea
2019 – Current	National Science and Engineering Undergraduate Scholarship by Korean Government - Student Aid Foundation - Full scholarship

Services & Other Professional Activities

2022	Undergraduate Student Council, UNIST - Election camp, Emergency Response Committee
2021	College of Information and Biotechnology Student Council, UNIST - Division of Academic Studies, School of Electrical Engineering
2020	School of Biomedical Engineering Student Council, UNIST - Preparation Task Force
2020	School of Electrical & Computer Science Engineering Student Council, UNIST - Emergency Response Committee - Planning department
2020	Campus Assistant, Freshman Support Center, UNIST - 1 Year Assistant for 18 Freshmen
2020	School of Human Factors Engineering Student Council, UNIST - Planning department